

Shock

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 Adapted from: [Shock](#)

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Learning Objectives (4)

Versions

After completing this brick, you will be able to:

- 1 Define shock.
- 2 Describe the clinical presentations of the different forms of shock.
- 3 Explain the physiologic parameters measured in shock.
- 4 Differentiate the five primary types of shock, including how they differ in cardiac output, afterload, and preload status.

CASE CONNECTION

ST, a middle-aged female, is brought into the emergency room after being involved in a high-speed car crash. The paramedic reports her blood pressure has been falling over the past 15 minutes, despite her receiving intravenous (IV)

fluid on the way in. On brief exam, She is hypotensive (BP 74/50 mm Hg), tachycardic (HR 120 bpm), with an enlarged, tender abdomen and cool extremities.

What treatment should ST receive in the emergency department? Consider your answer as you read, and we'll revisit ST at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Is Shock (LO1)?

Shock is a life-threatening condition characterized by inadequate delivery of oxygen to tissues (**hypoperfusion**) relative to their metabolic demands (**hypoxia**), resulting in cellular dysfunction and potential organ failure. This mismatch between oxygen supply and demand is not immediately fatal, but, depending on the degree of insufficiency, it can set off a cascade of cellular events that results in the death of the cell. Globally, shock can progress to multiorgan failure and permanent, often lethal organ damage.

While often associated with **hypotension** (typically systolic blood pressure <90 mmHg or mean arterial pressure <65 mmHg), shock can exist even with normal blood pressure readings during early compensatory phases. Also, individuals who normally have low resting blood pressures may tolerate lower blood pressures before succumbing to tissue hypoperfusion, whereas chronically hypertensive patients may experience the deleterious effects of hypoperfusion at higher-than-expected blood pressures. So, the assessment of shock requires both an objective measure of the vital signs and a more subjective assessment of organ perfusion.

blood pressure is normal.

Subjective assessment of organ perfusion indicates shock even if

Pathophysiology of Shock

The cellular oxygen imbalance of shock is most commonly related to impaired oxygen delivery in the setting of circulatory failure. When cells cannot obtain sufficient oxygen, aerobic metabolism fails, forcing a shift to anaerobic glycolysis. This less efficient pathway produces lactate and generates inadequate amounts of adenosine triphosphate (ATP), the cell's primary energy currency.

The resulting ATP deficit disrupts essential cellular functions, including ion pump activity (particularly Na^+/K^+ ATPase), leading to cellular swelling, calcium influx, activation of destructive enzymes, and ultimately cell death if the process continues unchecked.

Oxygen Delivery Deficit

The fundamental problem in shock is inadequate oxygen delivery (DO_2) to tissues. Oxygen delivery is determined by the product of cardiac output (CO) and arterial oxygen content (CaO_2): $DO_2 = CO \times CaO_2$.

oxygen delivery = cardiac output $\times$ arterial oxygen content

Cardiac output is the product of heart rate (HR) and stroke volume (SV), while stroke volume depends on preload, afterload (systemic vascular resistance or SVR), and myocardial contractility.

Types of Shock

Shock can be subdivided into five broad categories: distributive, cardiogenic, hypovolemic, neurogenic, and obstructive (Figure 1). These categories are based on the underlying pathophysiology of each and will be described in greater detail below.

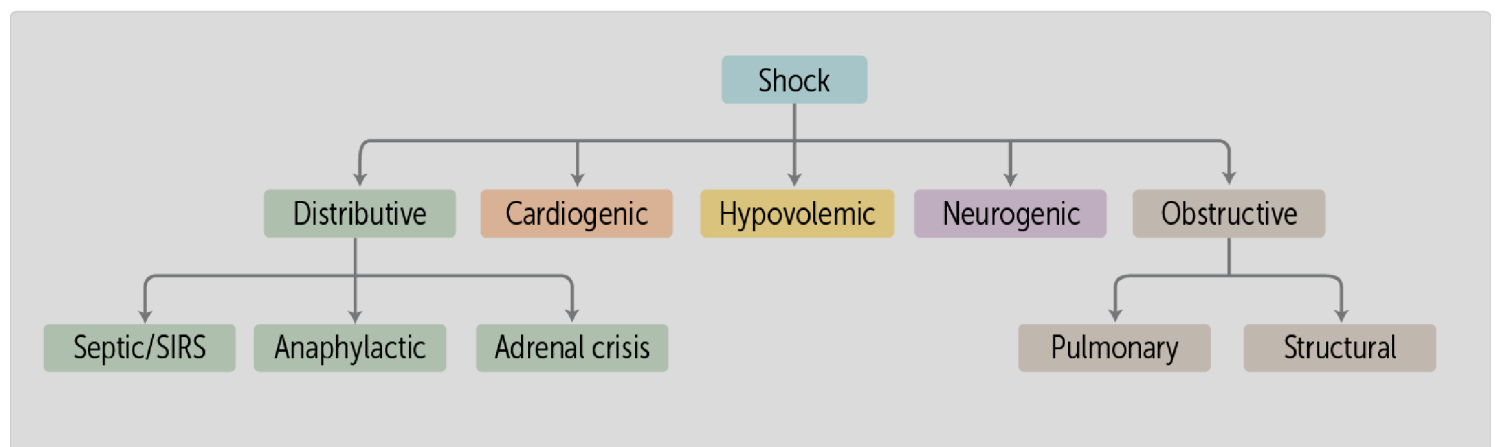


Figure 1 Types of shock

Clinical Presentation of Shock (LO2)?

The clinical presentation of shock combines common features across all types with distinctive characteristics that help identify the underlying mechanism. Early recognition of these patterns is crucial for directing appropriate diagnostic and therapeutic interventions.

Regardless of etiology, patients in shock typically exhibit several hallmark findings:

- **Hypotension:** Systolic blood pressure below 90 mmHg or mean arterial pressure below 65 mmHg (may be absent in early compensated shock). This is associated with syncope or presyncope, especially orthostatic.
- **Tachycardia** is nearly universal (except in certain conditions like neurogenic shock or in patients on beta-blockers) as the heart attempts to maintain cardiac output despite compromised stroke volume.
- **Altered mental status** reflects cerebral hypoperfusion and is a sensitive indicator of shock severity. It can range from anxiety and restlessness in early stages to confusion, lethargy, and eventually coma as cerebral perfusion deteriorates.
- Shortness of breath, with increased **respiratory rate** to compensate for metabolic acidosis.
- **Urine output decreases** as the kidneys experience reduced perfusion, with oliguria (output <0.5 mL/kg/hr) being a sensitive marker of shock.
- **Skin findings** often include cool, blue, or painful fingers and toes, though specific patterns vary by shock type.

Five types of shock: **N**ew **C**ells **D**own and **O**ut (from) **H**ypoxia

the metabolic demands of the tissues, resulting in tissue hypoxia.
Shock is a global reduction in blood flow that is insufficient to meet

Table 1 Summarizes the clinical findings seen in shock.

Table 1 Clinical findings seen in shock

System or location	Signs
Nervous	Mental status change (eg, confusion, loss of consciousness)
Cardiovascular	Sweating, ↑ heart rate, ↓ blood pressure
Respiratory	↑ Respiratory rate
Extremities	Cyanosis
Renal	↓ Urine output (oliguria)



output, cyanosis at extremities, and changes in mental status
Hypotension, tachycardia, tachypnea, sweating, decreased urine

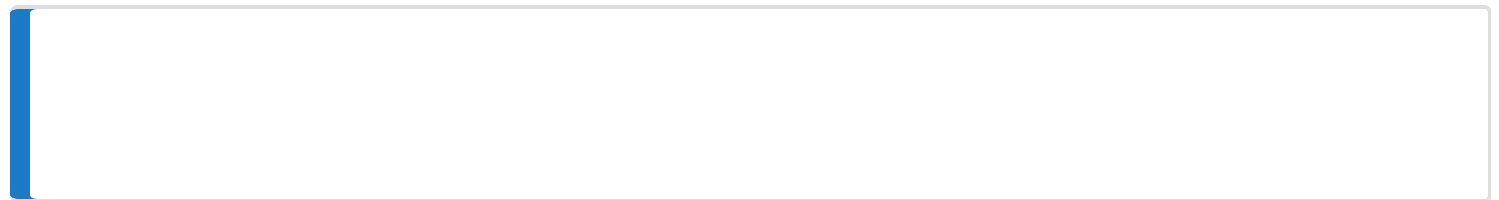


What Physiologic Parameters Do We Follow in a Patient with Shock (LO3)?

The assessment and management of shock rely on measuring key physiologic parameters that reflect cardiovascular function and tissue perfusion. These

measurements guide diagnosis, treatment decisions, and evaluation of therapeutic responses.

Mean arterial pressure (MAP) represents the average pressure driving blood flow through the systemic circulation during the cardiac cycle. These pressures are routinely measured by using a standard blood pressure cuff. It is calculated as below where DBP and SBP are diastolic and systolic blood pressure. Note pulse pressure can also be used and is (SBP-DBP).



A MAP of at least 65 mmHg is generally considered necessary for adequate perfusion of vital organs, though higher targets may be appropriate for patients with chronic hypertension.

Cardiac output (CO) is the volume of blood pumped by the heart per minute, normally 4-8 L/min. Cardiac output can be measured non-invasively through echocardiography or invasively via thermodilution with a pulmonary artery catheter.

$$\text{cardiac output} = \text{HR} \times \text{SV}$$

Stroke volume is decreased by:

- Decreased **preload**, which is a decrease in the pressure of the heart at the end of diastole and can be estimated by the end-diastolic volume (EDV)
- Increased **afterload**, which is an increase in the pressure resisting blood flow out of the heart and is usually determined by the SVR (arteriolar resistance) but also estimated by the mean arterial pressure
- Decreased **myocardial contractility**, which is a decrease in the strength of the heart muscle contractions and is independent of preload and afterload

Systemic vascular resistance (SVR) or total peripheral resistance (TPR) is the total resistance to blood flow and is mainly determined by arteriolar constriction, which is largely regulated by the sympathetic nervous system:

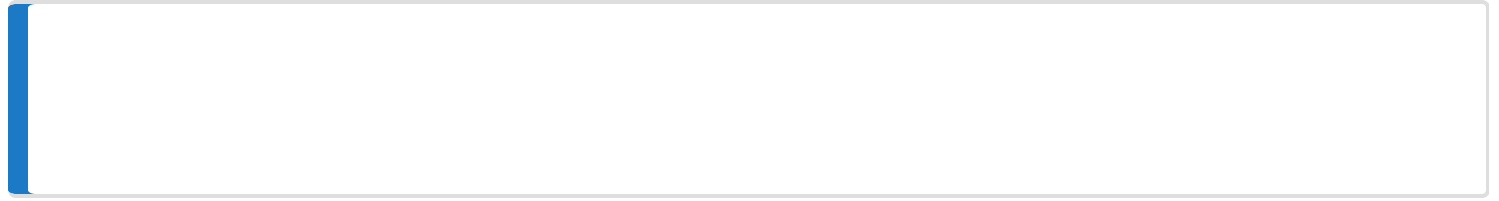
$$\text{SVR} = \text{MAP} / \text{cardiac output}$$

Arterial Oxygen Content

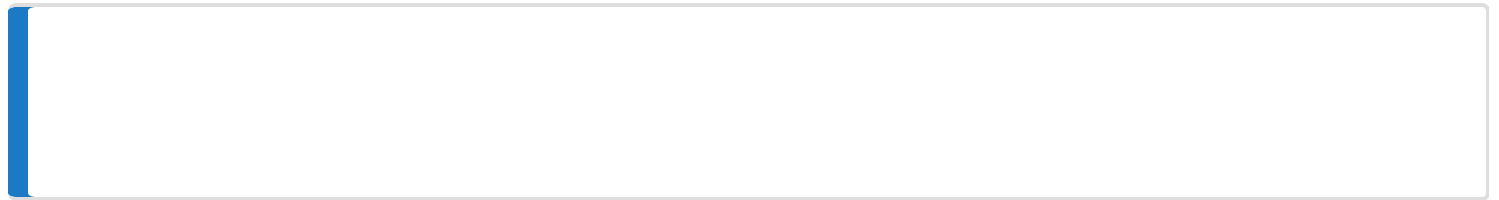
Because shock is ultimately caused by failed oxygenation of the blood or failed extraction of oxygen by the tissue, we often calculate the amount of oxygen delivered to the tissues. Oxygen delivery is determined by the product of cardiac output (CO) and arterial oxygen content (CaO_2): $\text{DO}_2 = \text{CO} \times \text{CaO}_2$.

Arterial oxygen content is primarily determined by hemoglobin concentration (Hb) and its O_2 carrying capacity of 1.36, oxygen saturation

(SaO_2), and to a lesser extent, dissolved oxygen (PaO_2) and its solubility of 0.03. This relationship is expressed as below:



The latter term of dissolved oxygen in the blood is very small, so the content equation can be simplified as:



C_aO_2 and $(\text{SaO}_2)_a$ are measured in millimeters of mercury (mm Hg). The $(\text{SaO}_2)_a$ is measured using routine arterial blood gas analysis, and the Hb is measured as part of the complete blood count.

Arterial oxygen content could fall because of:

- Decreased cardiac output
- Anemia (decreased Hb)
- Decreased oxygen saturation of arterial blood (hypoxemia), usually due to lung disease

Sometimes, the oxygen content is measured in both arterial and venous blood, and the difference is calculated (termed the **arteriovenous oxygen**

difference). This value is sometimes useful diagnostically. For example, in septic shock, the tissues often do not extract much oxygen, making for a small difference between arterial and venous oxygen content (small arteriovenous oxygen difference).

What Are the Five Types of Shock (LO4)?

Shock can be classified into five primary types based on the underlying pathophysiologic mechanism.

Each type presents with a characteristic pattern of alterations in cardiac output (CO), systemic vascular resistance (afterload), and preload (EV) that guides both diagnosis and management strategies ([Figure 3](#)). Each of these types of shock are described in detail in the sections below.

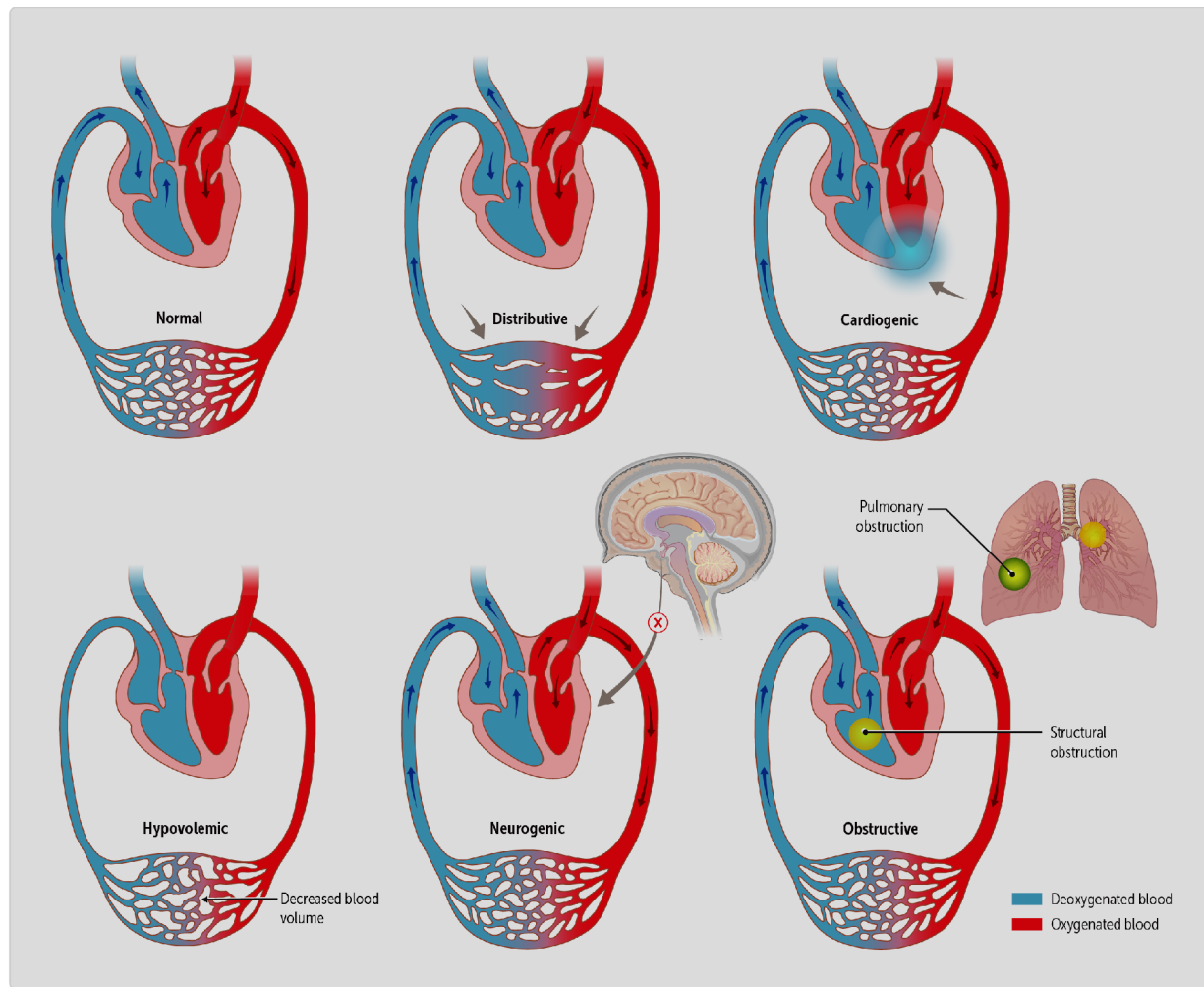


Figure 3 Normal state and trigger insults of the five types of shock

Distributive Shock: Decreased Systemic Vascular Resistance (blood pooling)

Distributive shock results from pathologic vasodilation and increased vascular permeability, leading to maldistribution of blood flow and relative hypovolemia. In distributive shock, extensive systemic vasodilation causes a decrease in SVR and blood pressure.

The primary culprit in distributive shock is **low SVR** due to vasodilation. The low SVR forces the body to compensate and increase the cardiac output in an attempt to maintain the MAP. The body increases cardiac output by increasing both heart rate (tachycardia) and stroke volume, each driven by

the sympathetic nervous system. Blood is pumped rapidly out of the heart, leading to a decreased EDV (preload)

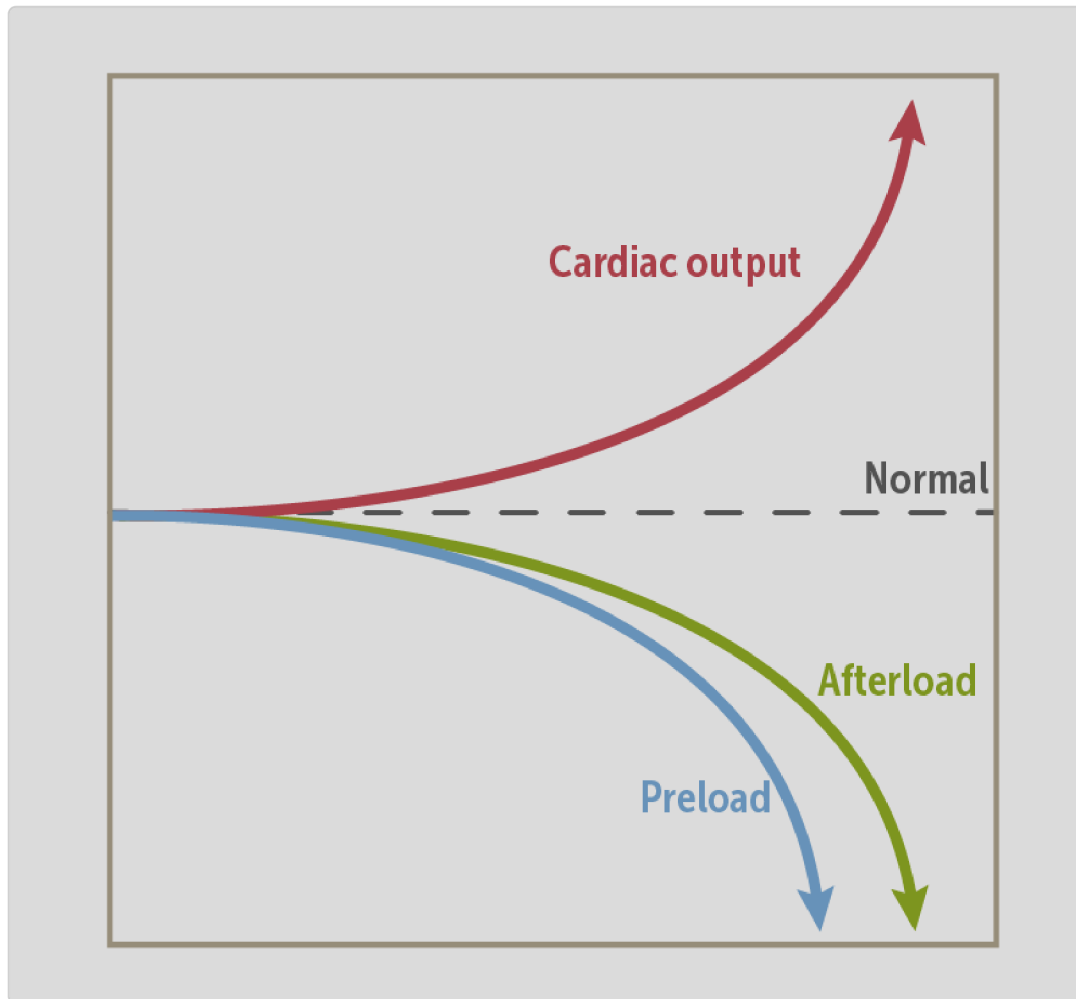


Figure 4 Distributive shock

What causes this vasodilation? There is a wide range of causes:

- **Septic shock**, the most common form of distributive shock, results from a bacterial infection leading to life-threatening organ dysfunction. Microbial components, including lipopolysaccharides, stimulate inflammatory cells to produce cytokines such as transforming growth factor (TGF), interleukin-1 (IL-1), interferon γ , IL-6, IL-12, and IL-18. These cytokines mediate endothelial cell damage and the release of

vasodilators such as nitric oxide and prostaglandin I₂ resulting in microcirculatory abnormalities. Septic shock patients often have fever, tachypnea, and a clear infectious source

- **Systemic inflammatory response syndrome (SIRS)**: This is a complex inflammatory response to stress that involves the entire body. The presentations vary, and different patients may have tachycardia, hypothermia, hyperthermia, and peripheral WBC counts that are very low or very high. SIRS may lead to the failure of multiple organs (kidney, heart, and lungs). Causes can include sepsis but can also include noninfectious inflammation due to trauma, pancreatitis, or burns. There is activation of humoral and cellular immunity, the complement system, and cytokines.
- **Anaphylactic shock**: This is reexposure to an antigen against which an individual already has produced specific IgE antibodies. This leads to vasodilation through basophil degranulation and release of histamine, prostaglandins, and leukotrienes. Anaphylactic shock presents with urticaria, angioedema, bronchospasm, and a history of allergen exposure
- **Adrenal crisis**: This is a major deficiency of cortisol and/or aldosterone, often due to abrupt withdrawal of chronically administered corticosteroid drugs. These hormones are critical in maintaining the SVR, and there may be hypotension or shock when they are profoundly decreased. adrenal crisis may present with hyperpigmentation, hyponatremia, and hyperkalemia.

Clinically, the patient will be hypotensive, but the peripheral vasodilation may cause the skin to be warm even at the extremities, which are usually cold in the other types of shock.

leading to decreased SVR.

MAP will be low during distributive shock because of vasodilation.

Cardiogenic Shock: Pump failure

In cardiogenic shock, blood pressure drops because of the heart's failure to pump blood through the circulatory system (**decreased cardiac output**). Some of the causes can include myocardial ischemia and infarction, cardiomyopathies, valvular disease, and arrhythmias.

To maintain blood pressure in the face of low cardiac output, the body must raise the SVR as a compensatory response. This arteriolar constriction leads to high afterload in cardiogenic shock. The high SVR exists despite the low blood pressure, which can be clinically confusing! Just remember that blood pressure depends not only on SVR but a combination of SVR, cardiac output, and blood volume; here, the cardiac output is very low and dominates the clinical picture (**Figure 5**).

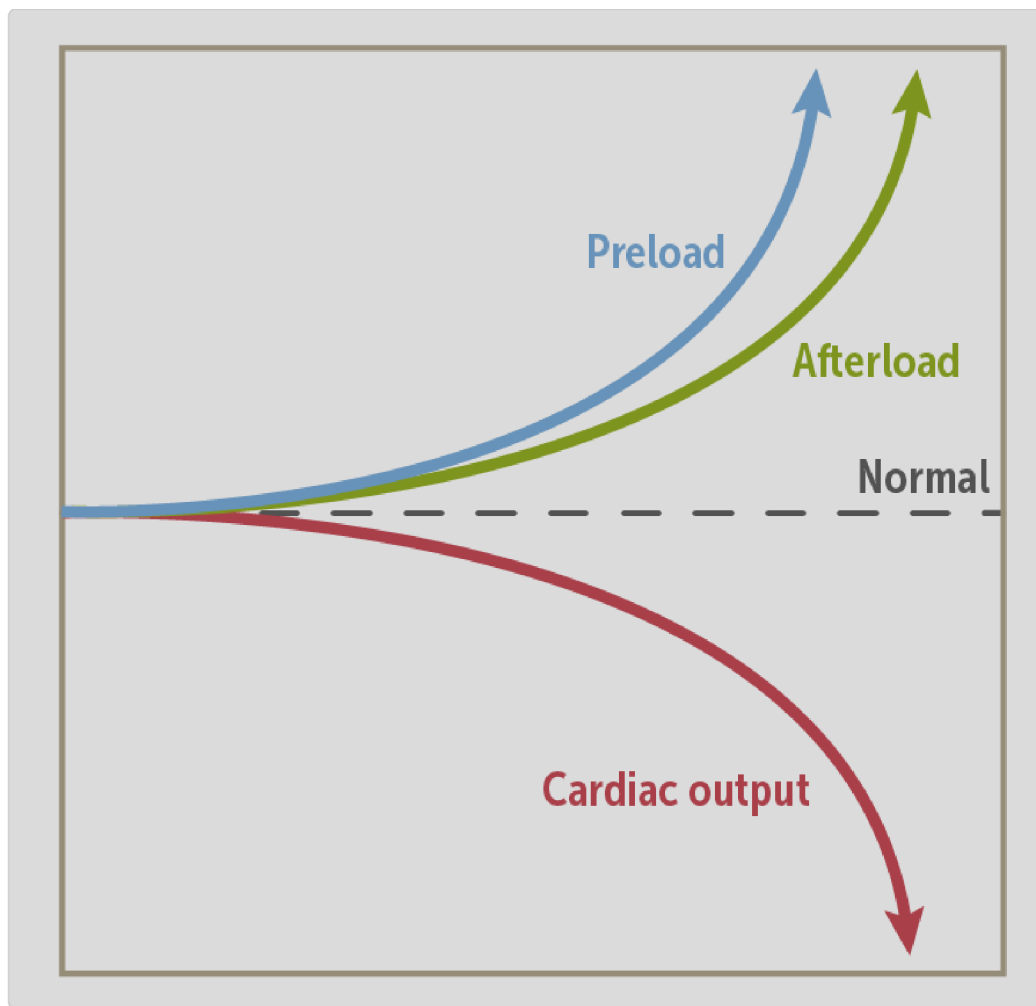


Figure 5 Cardiogenic shock

The pathophysiology involves a vicious cycle where reduced cardiac output leads to coronary hypoperfusion, worsening myocardial ischemia and further decreasing contractility. Compensatory mechanisms, including sympathetic activation and renin-angiotensin-aldosterone system activation, initially support blood pressure but ultimately increase myocardial oxygen demand and afterload, exacerbating cardiac dysfunction.

Clinically, the patient has cool extremities, tachycardia, weak pulses, elevated JVD, and an S3 heart sound. The EDV (preload) rises due to the inability of the heart to adequately pump the blood, leading to pulmonary

edema. Pulmonary edema results in dyspnea, crackles and hypoxemia (from pulmonary edema),



CLINICAL CORRELATION

To treat cardiogenic shock, one possibility is an inotropic drug to increase cardiac contractility, although these generally also increase the risk of dangerous arrhythmias. Vasodilators decrease systemic vascular resistance and, in theory, should decrease the amount of work demanded by the failing heart. However, the blood supply to the heart itself requires adequate systemic vascular resistance to sustain the diastolic blood pressure, since the coronary arteries only fill during diastole. Therefore, vasodilators should not be used to treat cardiogenic shock in isolation; some drugs like dobutamine and milrinone act as both inotropes and vasodilators but still run the risk of causing decompensation of the blood pressure in a patient already in shock.

infarction, cardiomyopathies, valvular disease, and arrhythmias. Cardiogenic shock may result from myocardial ischemia and

Hypovolemic Shock: Decreased Preload

Hypovolemic shock is caused by significant loss of blood or plasma volume, most often by a sudden **hemorrhage**. Far less often, it can result from routine causes of dehydration, like overuse of diuretics, severe vomiting or diarrhea, or lack of fluid intake. It can also result from the shifting of fluid from the vasculature to body cavities such as during pancreatitis, burns, or peritonitis. This is sometimes referred to as internal third-spacing.

In hypovolemic shock, less volume circulating in blood vessels means lower venous return, reduced ventricular filling, and reduced preload. The reduced preload leads to reduced stroke volume, which reduces cardiac output. SVR increases as a compensatory mechanism to maintain blood pressure via arteriolar constriction.

The body compensates for hypovolemia through multiple mechanisms: baroreceptor-mediated sympathetic activation increases heart rate and contractility while causing vasoconstriction in non-essential vascular beds (skin, splanchnic circulation, kidneys); the renin-angiotensin-aldosterone system promotes sodium and water retention; and antidiuretic hormone release further enhances water conservation. These mechanisms can temporarily maintain blood pressure despite significant volume loss, masking the severity of shock until decompensation occurs. However, because of low preload, the cardiac output is generally low (**Figure 6**).

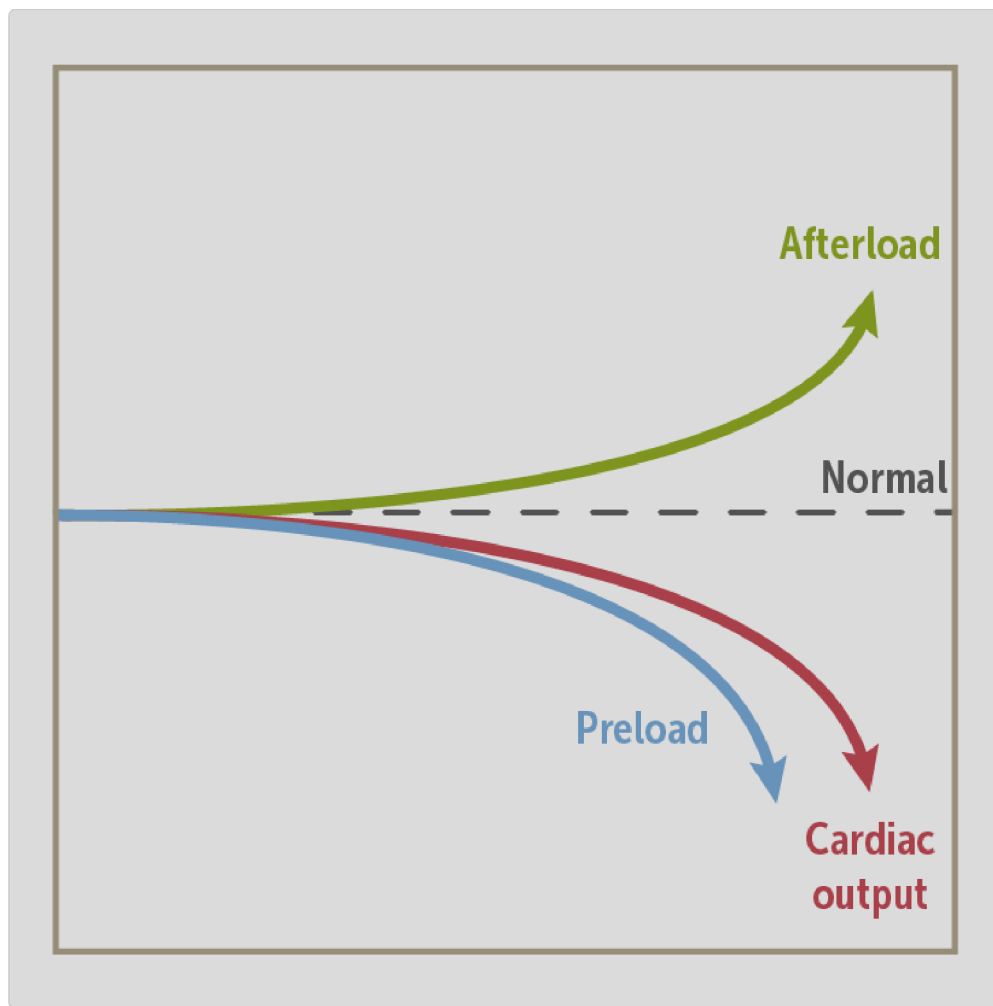


Figure 6 Hypovolemic shock

Clinically, the patient has cold extremities (from vasoconstriction) and may appear pale due to the hemorrhage. A source of bleeding (trauma, GI) is often obvious. The severity of hemorrhagic shock correlates with the volume of blood lost, with clinical manifestations typically appearing after 15-30% of blood volume is depleted

The low blood volume reduces the preload and stroke volume.

Neurogenic Shock: Everything Down

Neurogenic shock results from disruption of sympathetic nervous system control, typically following high spinal cord injury (above T6) or brain injury. The loss of sympathetic tone causes widespread vasodilation (decreased SVR) and loss of cardiac sympathetic stimulation, often resulting in bradycardia rather than the tachycardia typical of other shock states.

The failure of the autonomic system disrupts everything, preventing compensatory responses. There is concurrent loss of sympathetic stimulation, which reduces arterial tone, leading to vasodilation (low SVR and afterload), low heart rate and contractility (low cardiac output), and low venous constriction (low EDV and preload). Remember, this is the one with low everything (**Figure 7**).

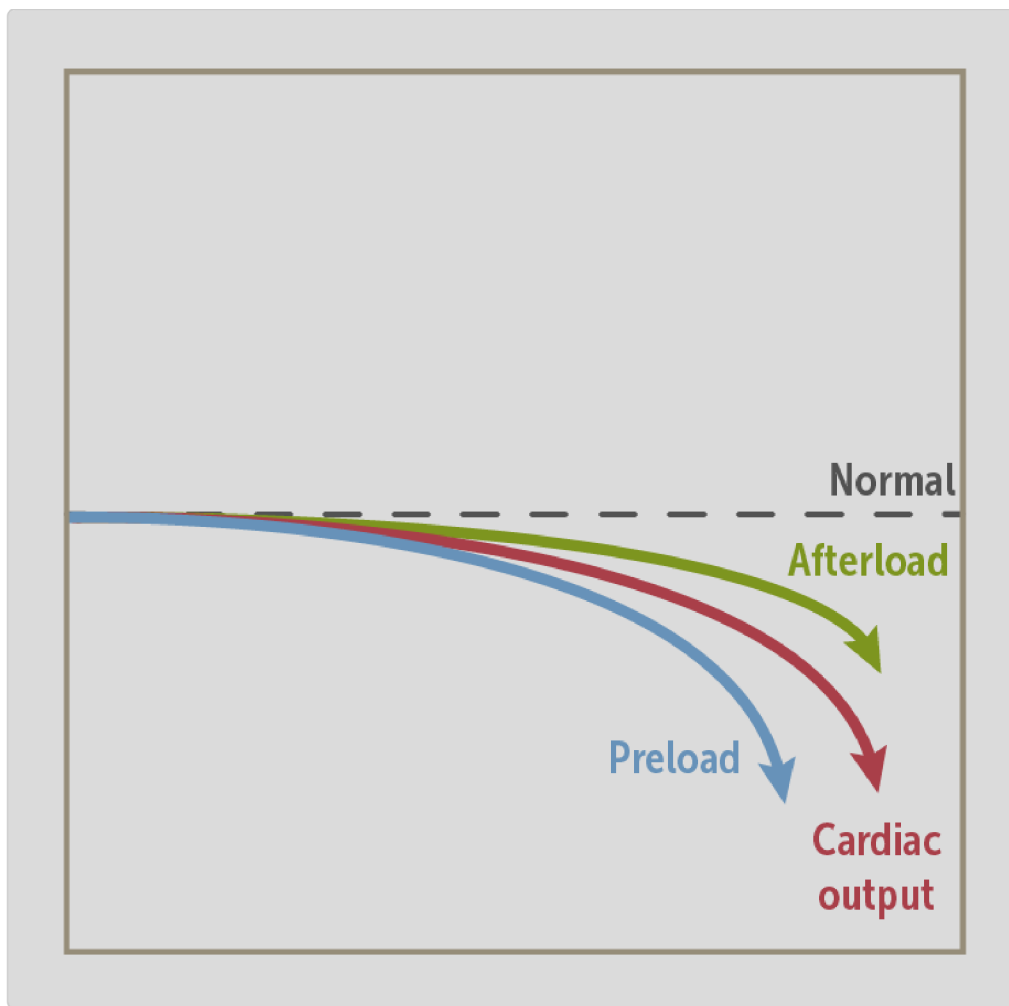


Figure 7 Neurogenic shock

Clinically, neurologic signs and symptoms and a history of trauma to the brain or spinal cord are often seen. The classic triad includes hypotension, bradycardia (instead of the expected tachycardia), and warm, dry skin below the level of injury due to loss of sympathetic tone. It's also the only type of shock in which the patient does not have tachycardia since the sympathetic nervous system cannot mount that response. Some authors place this under the umbrella of distributive shock since the SVR drops, but the cardiac response is quite different.

None, since the autonomic nervous system is not functioning

Obstructive Shock: Obstruction Causing Decreased Cardiac Output

Obstructive shock is a blockage cardiac filling or emptying via mechanical obstruction, despite normal heart function and blood volume. It is a little more complex because there are two subtypes, each of which decreases MAP by a slightly different mechanism ([Figure 8](#)).

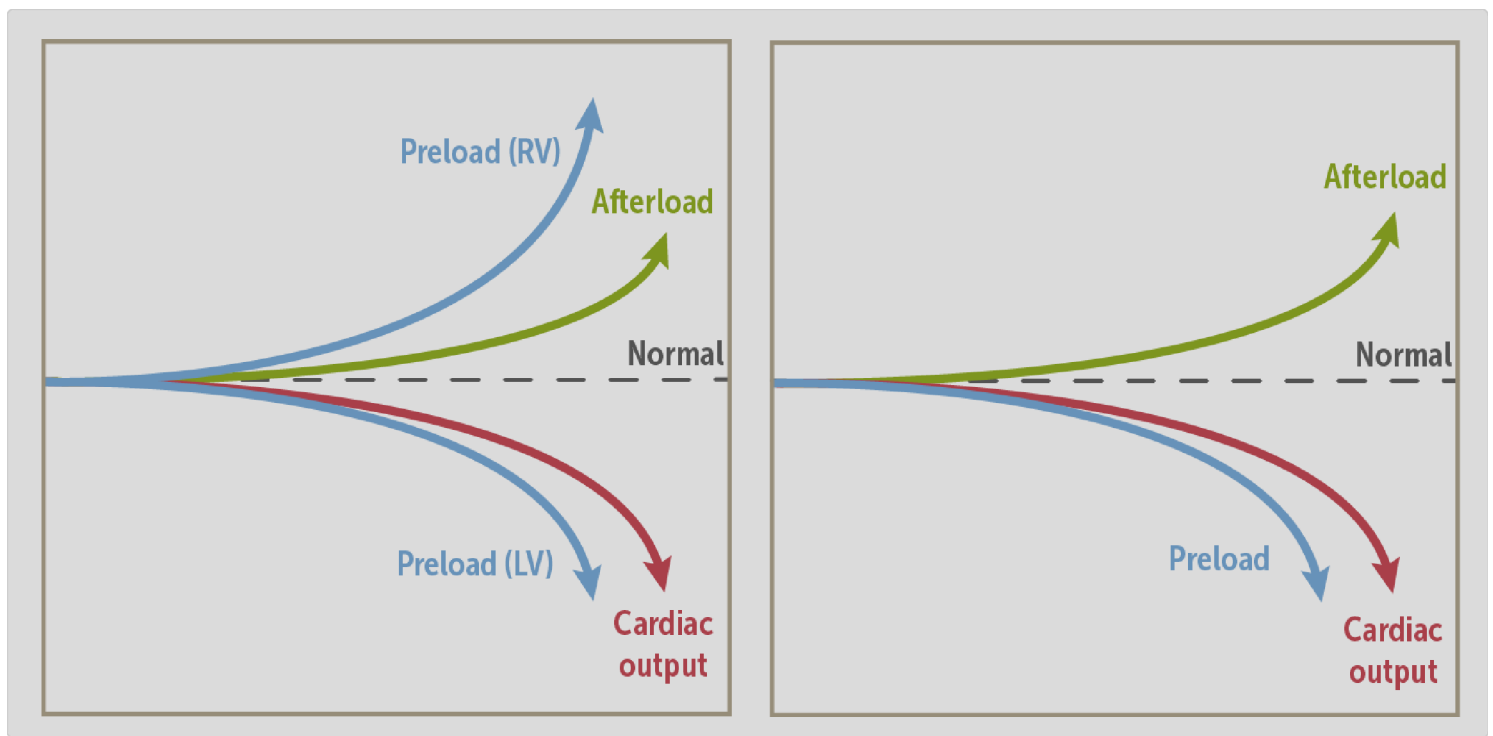


Figure 8 Obstructive shock: pulmonary (left) and structural (right)

- In **pulmonary obstructive shock** (Figure 8, left), increased resistance to flow within the pulmonary circulation prevents blood from reaching the left side of the heart. This increases right ventricular preload but reduces the left ventricular preload (and stroke volume), causing total cardiac output to drop. The most common example is a **massive pulmonary embolism**.
- In **structural obstructive shock** (Figure 8, right), the resistance is caused within the heart itself, which is restricted from filling with the necessary amount of blood. This decreases right and left ventricle preload and cardiac output. Examples are **tension pneumothorax**, or **cardiac tamponade**.

In both cases, cardiac output is low because of increased resistance in some parts of the vascular system and poor filling of the left ventricle. The body attempts to compensate by increasing heart rate to try to increase the

cardiac output. The SVR (afterload) increases in response to the low cardiac output. So far, this looks much like cardiogenic shock. However, unlike in cardiogenic shock, in obstructive shock, the preload in the LV is decreased because the obstruction prevents venous return.

Clinically, the most obvious distinguishing features are the signs and symptoms of the underlying cause: leg swelling, dyspnea, and pleuritic chest pain in pulmonary embolism; absent breath sounds in pneumothorax; or distant heart sounds and jugular venous distension in cardiac tamponade.

reduced.

Less blood fills the left ventricle during diastole, so cardiac output is

Comparison of Shock Types

Can you identify which type of shock a patient might have from their MAP, SVR, EDV, cardiac output, and heart rate? In [Table 2](#), the precipitating cause(s) of hypotension for each type of shock is shown with a P. See if you can quiz yourself to identify the precipitating cause and the compensatory mechanisms. An asterisk highlights something unique to that column.

Table 2 Physiologic characteristics for each type of shock

	MAP	EDV (preload)	SVR	Cardiac output	Heart rate
Distributive	↓	↓	↓P	*↑	↑
Cardiogenic	↓	*↑	↑	↓P	↑
Hypovolemic	↓	↓P	↑	↓	↑
Neurogenic	↓P	↓P	↓P	↓P	*Normal or ↓
Obstructive: pulmonary	↓	*↑ (RV) ↓ (LV)	↑	↓P	↑
Obstructive: structural	↓	↓	↑	↓P	↑

Prognosis

What outcome can we expect for a patient in shock? Shock is a highly critical state. Mortality rates are 25% to 50% for cardiogenic and distributive shock but lower for the other types.



CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

Thinking back to ST, what treatment should be initiated in the emergency department?

ST is in hypovolemic shock because of low cardiac output due to blood loss into her abdomen (hemoperitoneum). Because the preload must be increased, the rate of her IV fluids is accelerated, and a blood transfusion is arranged. Surgery is urgently consulted to manage her hemoperitoneum. ST is taken to the operating room, her internal bleeding is controlled, and she makes a successful recovery.

Summary

- Shock is a life-threatening condition characterized by inadequate oxygen delivery to tissues, leading to cellular dysfunction and potential organ failure.
- Shock can be subdivided into five categories: distributive, cardiogenic, hypovolemic, neurogenic, and obstructive, each with distinct pathophysiological mechanisms.
- Clinical presentation of shock includes hypotension, tachycardia, altered mental status, shortness of breath, decreased urine output, and skin changes.
- Shock progresses through stages: compensated, uncompensated, and end-organ failure, with early recognition and intervention being critical.
- Key physiologic parameters to monitor in shock include mean arterial pressure, cardiac output, stroke volume, and systemic vascular resistance.
- Treatment of shock is tailored to the type and underlying cause, with initial stabilization often involving fluid resuscitation if pulmonary edema is absent.

Review Questions

1. A 60-year-old male has a myocardial infarction. He is initially stabilized but develops a ventricular arrhythmia hours later and loses consciousness. His blood pressure falls to 90/60 mm Hg, pulse is 130 beats/min, and respirations are 20/min. His exam shows tachycardia. Lungs are clear. A change in which of the following parameters contributes most to this patient's hypotension?

- A. Arterial oxygen content
- B. Contractility
- C. Heart rate
- D. Preload
- E. Vascular radius (r)

Explanation (requires correct answer)

2. A 52-year-old truck driver suffers a pulmonary embolism after developing a deep venous thromboembolism during a 10-hour drive. If he develops shock from the pulmonary embolism, which of the following will most likely be seen?

- A. Decreased contractility
- B. Decreased heart rate
- C. Decreased systemic vascular resistance
- D. Increased end-diastolic volume
- E. Increased systemic vascular resistance

Explanation (requires correct answer)

3. A 72-year-old female develops septicemia and goes into shock after abdominal surgery. Which of the following symptoms raises the greatest concern for end-organ failure?

- A. Anuria
- B. Blood pressure of 90/65 mm Hg
- C. Heart rate of 120/min
- D. Respiration rate of 26/min
- E. Temperature of 101°F

Explanation (requires correct answer)

pattern of shock do you expect in this patient.

- A. Cardiogenic shock
- B. Distributive shock
- C. Hypovolemic shock
- D. Neurogenic shock
- E. Obstructive shock

Explanation (requires correct answer)

5. In **cardiogenic shock**, which of the following physiologic changes is most accurate?

- A. MAP ↑, EDV ↓, SVR ↓, CO ↑
- B. MAP ↓, EDV ↑, SVR ↑, CO ↓

C. MAP ↓, EDV ↓, SVR ↑, CO ↑

D. MAP ↑, EDV ↑, SVR ↓, CO ↓

Explanation (requires correct answer)

How would you rate this Brick?



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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- | | |
|----|--|
| 01 | Define shock. |
| 02 | Describe the clinical presentations of the different forms of shock. |
| 03 | Explain the physiologic parameters measured in shock. |
| 04 | Differentiate the five primary types of shock, including how they differ in cardiac output, afterload, and preload status. |

Objective Confidence: 0/4